



BobSim

Ramp-Steer Vehicle Characterization

Measured- a_y ramp-response isolines with robust metric fits

Generated: 2026-06-23 03:34

Run Notes

- Open-loop constant-rate handwheel ramp at the configured test velocities
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis
- Metrics describe the ramp-response path, not fully settled equilibrium holds
- $|a_y| < 1.0 \text{ m/s}^2$ is excluded; linear band = 1.0 to 4.0 m/s^2
- Ramp terminates when local lateral gain drops 20% from the linear reference
- Limit metrics are local derivatives at the largest measured $|a_y|$ reached
- Tire-lift protection terminates if any tire falls below the configured normal-load threshold
- Raw plots are rendered; summary metrics use robust smoothed derivatives

RampSteerEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	1.01 → 16.37	m/s^2
Linear Sideslip Gradient	0.00084	$\frac{\text{rad}}{\text{m/s}^2}$
	0.47	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	−0.00355	$\frac{\text{rad}}{\text{m/s}^2}$
	−2.00	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00061	$\frac{\text{rad}}{\text{m/s}^2}$
	0.34	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.01873	$\frac{\text{rad}}{\text{m/s}^2}$
	10.53	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.03082	$\frac{\text{rad}}{\text{m/s}^2}$
	17.32	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.03121	$\frac{\text{rad}}{\text{m/s}^2}$
	17.54	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.889	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.954	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-20.9	$\text{N} \cdot \text{m}$
Peak Torque a_y	16.24	m/s^2

RampSteerEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	1.01 $\rightarrow$ 16.37	m/s^2
Linear Sideslip Gradient	0.00084	$\frac{\text{rad}}{\text{m/s}^2}$
	0.47	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	-0.00355	$\frac{\text{rad}}{\text{m/s}^2}$
	-2.00	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00061	$\frac{\text{rad}}{\text{m/s}^2}$
	0.34	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.01873	$\frac{\text{rad}}{\text{m/s}^2}$
	10.53	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.03082	$\frac{\text{rad}}{\text{m/s}^2}$
	17.32	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.03121	$\frac{\text{rad}}{\text{m/s}^2}$
	17.54	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.889	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.954	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-20.9	$\text{N} \cdot \text{m}$
Peak Torque a_y	16.24	m/s^2

RampSteerEval Summary 17.5 m/s

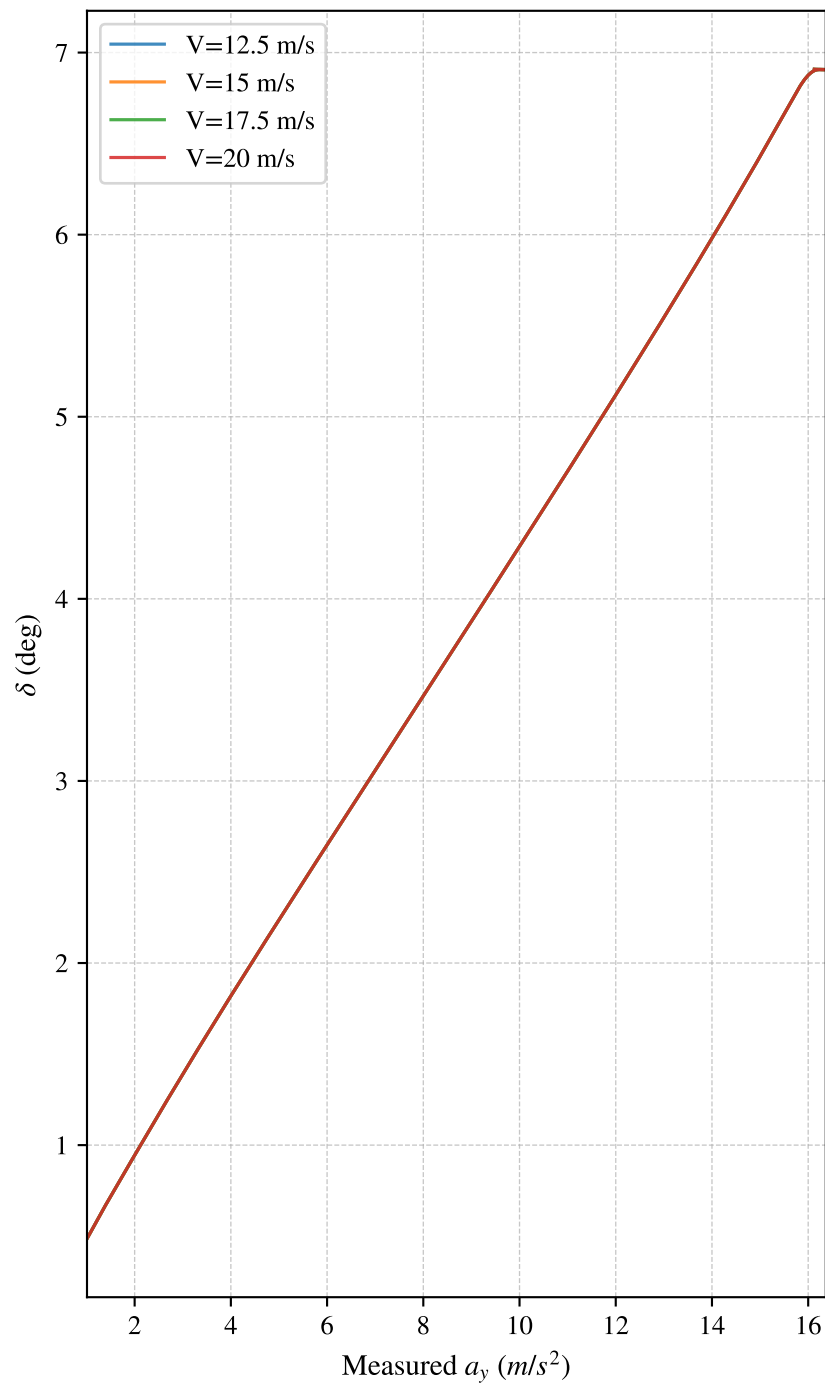
Metric	Value	Units
Measured a_y Range	1.01 $\rightarrow$ 16.37	m/s^2
Linear Sideslip Gradient	0.00084	$\frac{\text{rad}}{\text{m/s}^2}$
	0.47	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	-0.00355	$\frac{\text{rad}}{\text{m/s}^2}$
	-2.00	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00061	$\frac{\text{rad}}{\text{m/s}^2}$
	0.34	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.01873	$\frac{\text{rad}}{\text{m/s}^2}$
	10.53	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.03082	$\frac{\text{rad}}{\text{m/s}^2}$
	17.32	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.03121	$\frac{\text{rad}}{\text{m/s}^2}$
	17.54	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.889	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.954	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-20.9	$\text{N} \cdot \text{m}$
Peak Torque a_y	16.24	m/s^2

RampSteerEval Summary 20.0 m/s

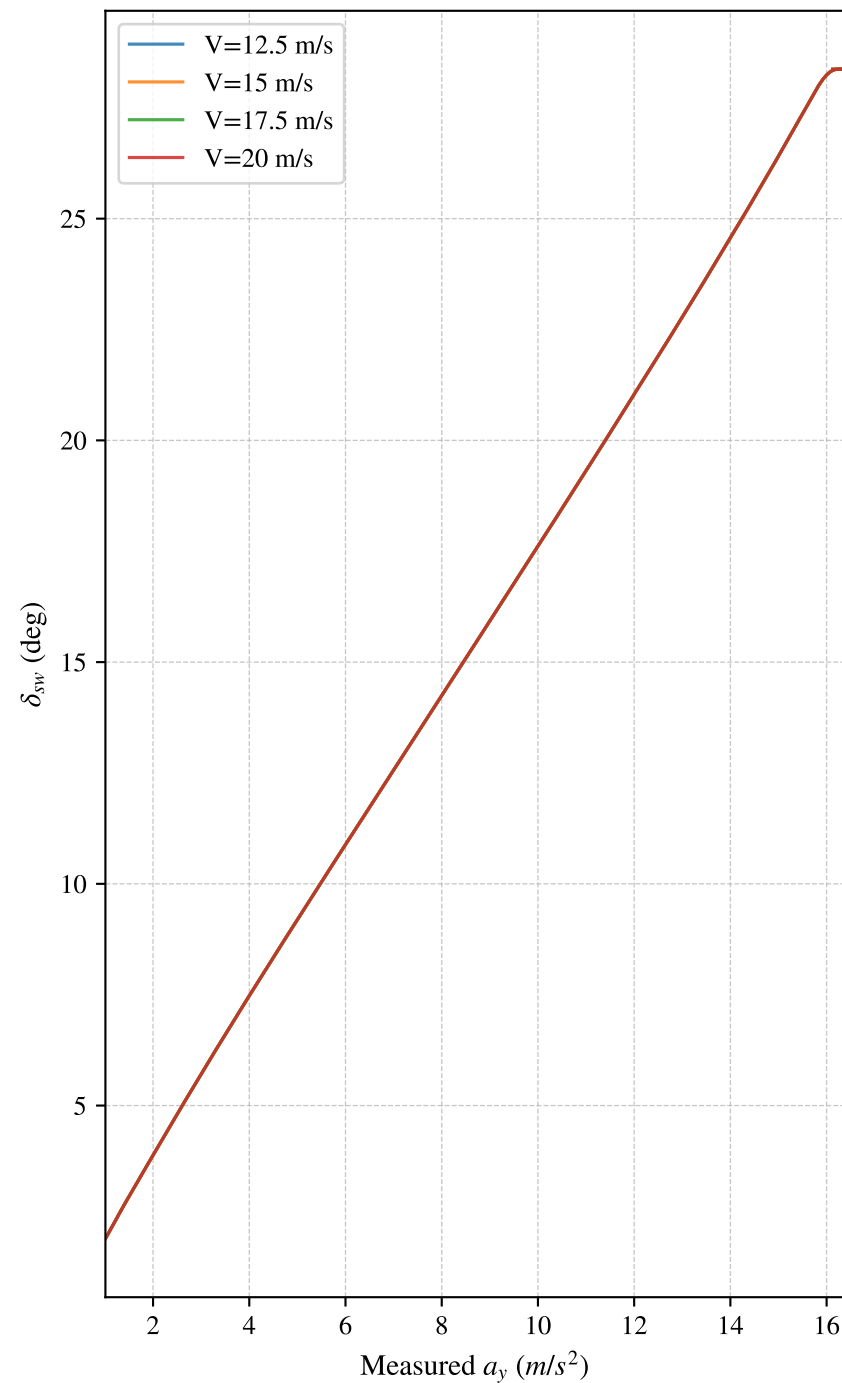
Metric	Value	Units
Measured a_y Range	1.01 $\rightarrow$ 16.37	m/s^2
Linear Sideslip Gradient	0.00084	$\frac{\text{rad}}{\text{m/s}^2}$
	0.47	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	-0.00355	$\frac{\text{rad}}{\text{m/s}^2}$
	-2.00	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00061	$\frac{\text{rad}}{\text{m/s}^2}$
	0.34	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	0.01873	$\frac{\text{rad}}{\text{m/s}^2}$
	10.53	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.03082	$\frac{\text{rad}}{\text{m/s}^2}$
	17.32	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.03121	$\frac{\text{rad}}{\text{m/s}^2}$
	17.54	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.889	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.954	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-20.9	$\text{N} \cdot \text{m}$
Peak Torque a_y	16.24	m/s^2

Raw Steering Samples vs Measured a_y

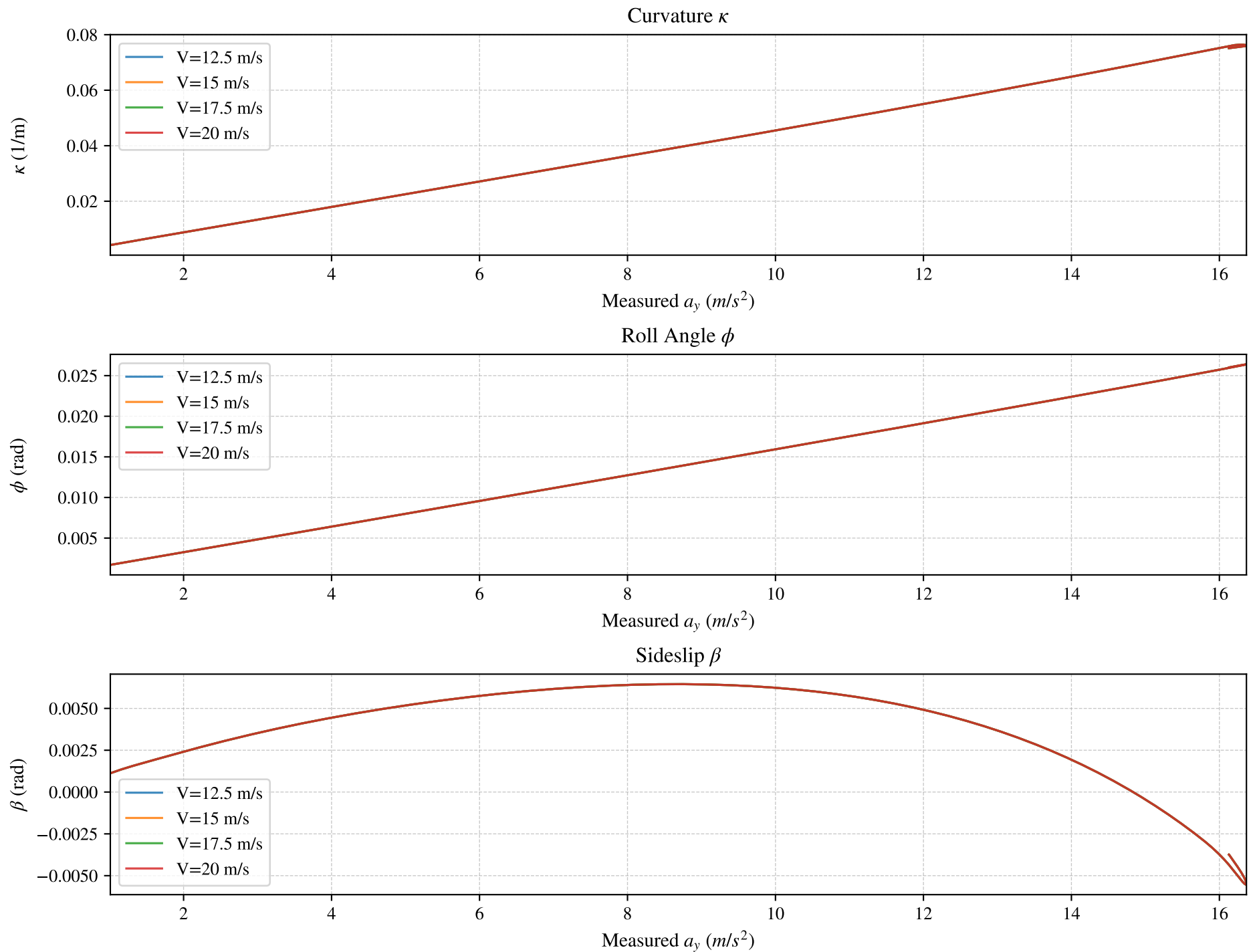
Roadwheel Angle δ



Handwheel Angle δ_{sw}

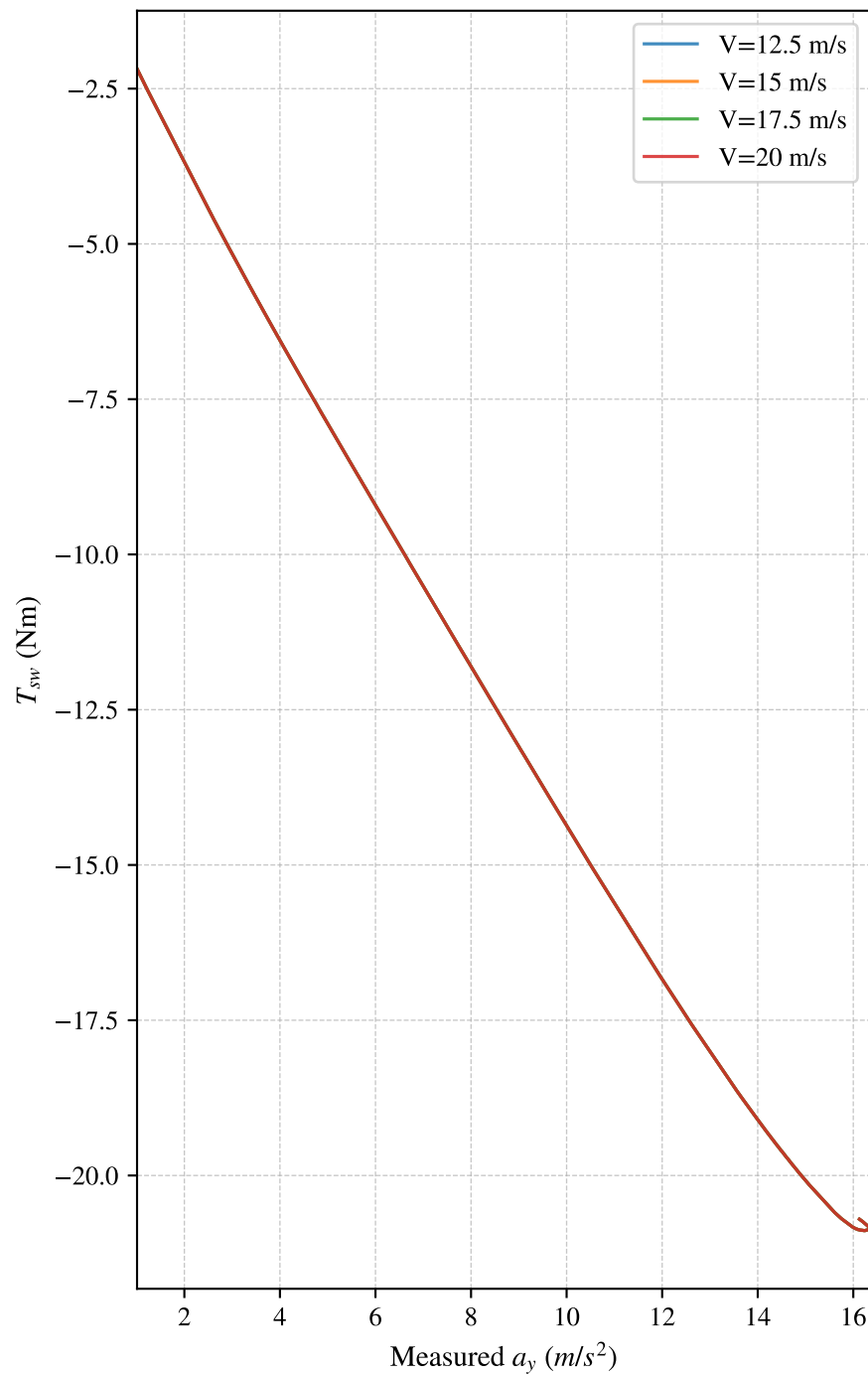


Raw Vehicle State Samples vs Measured a_y



Raw Steering Effort Samples vs Measured a_y

Steering Torque T_{sw}



Steer Excess δ_{excess}

